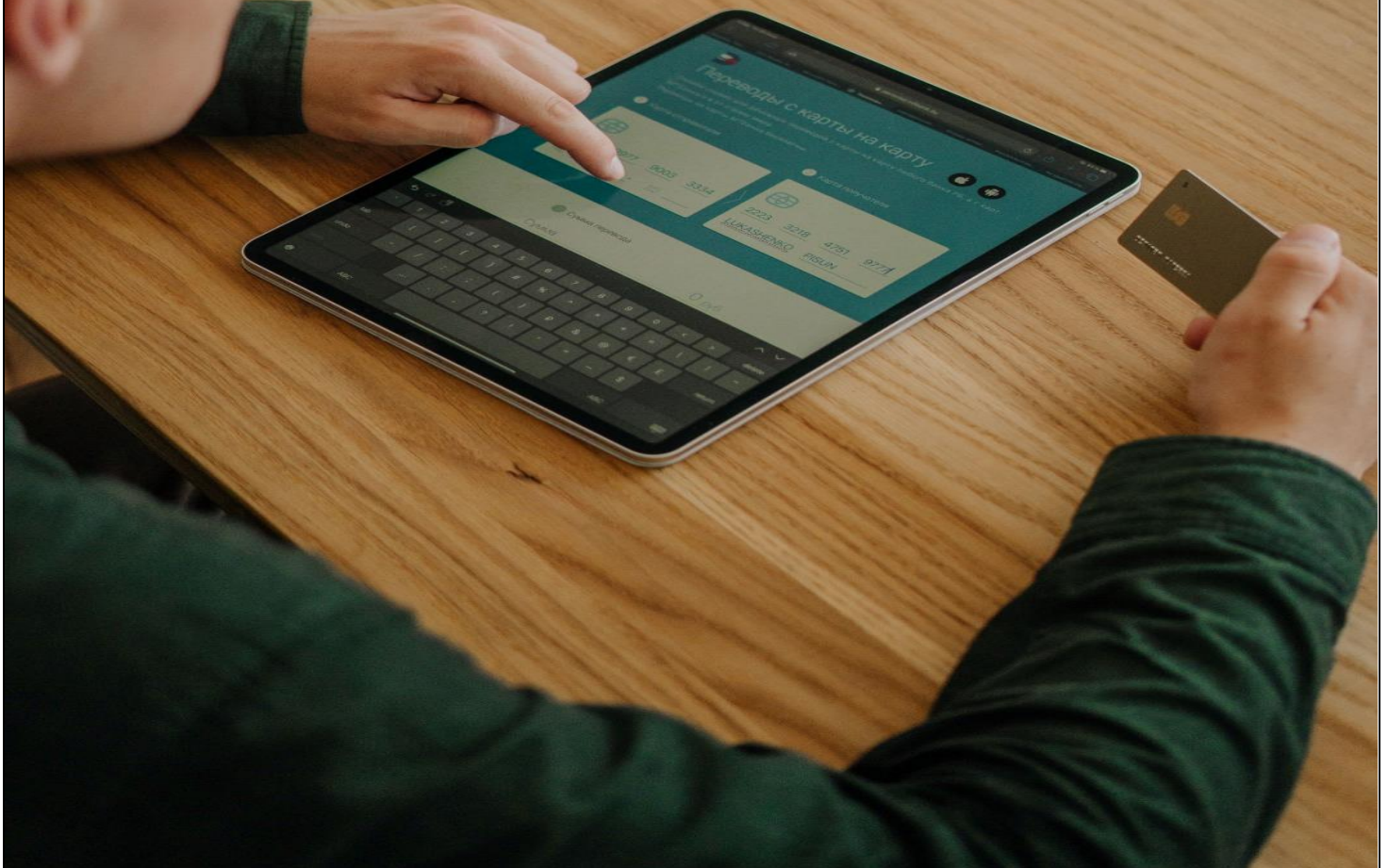


AssureX Warranty Claim Engine



PROJECT TITLE: ASSUREX CLAIM ENGINE

COMPETITION: TECHWIZ 7

Category & Theme: NextWave AI and ML / AI-Powered Document

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PROBLEM DEFINITION:

Insurance Claim Processing is often a time-consuming and complex process. Traditional Claim Processing methods rely heavily on manual work so This AssureX Claim engine is proposed to address these challenges by providing an intelligent and automated Claim processing system.

BACKGROUND AND BUSINESS NECESSITY:

AssureX Claim Engine is designed as an AI-based claim processing solution that automates important stages of the insurance claim lifecycle. The business necessity of AssureX arises from the need to make insurance claim processing faster, more accurate, cost-effective and scalable.

PROPOSED SOLUTION:

It proposes an AI-powered automated system that streamlines the insurance claim process by using AI, ML, OCR and NLP to extract and analyze claim information. It reduces manual effort and improves accuracy. It supports faster and more consistent claim decisions.

PURPOSE OF THE DOCUMENT:

The purpose is to define and explain the AssureX Claim Engine, its objectives, functionality and proposed solution.

SCOPE OF THE PROJECT:

- Automated Claim Submission
- Document Processing
- Data Validation
- AI/ML based claim Assessment
- Fraud Detection
- NLP integration
- Claim Classification
- Claim Tracking
- Decision Support
- Data Security
- Human Review

ASSUMPTIONS:

It assumes that the information provided by users is accurate, complete and valid. It is assumed that the required documents are available. The system uses predefined business rules, AI-based predictions and available policy information to support claim assessment and decision making.

CONSTRAINTS:

The AssureX Claim Engine may be constrained by the quality, availability and accuracy of claim and policy data. The system's AI and ML predictions may not be 100% accurate, especially for complex or unusual claims. Its performance may also depend on the availability of reliable data bases, OCR and NLP technologies and limited access to authorized information can restrict the system functionally.

FUNCTIONAL REQUIREMENTS:

User Management: The system will provide user authentication.

Product and Warranty Management: Users will be able to register products and manage warranty details.

Document & Evidence Management: The system allows users to upload product images and damage evidence.

OCR & Data Extraction: The system will extract important information from receipts and invoices.

Data Validation & Pre-Processing: It will validate submitted information, handle missing or incorrect data and prepare claim data for further processing using python.

Claim Registration & Management: Users and service center employees will be able to create and submit warranty claims with unique claim ID and track their claim status.

Claim Classification using AI/ML: The system shall use a Python-based machine learning model to classify claims as Valid claim, Invalid Claim or Manual Review and generate confidence score.

Claim Summary & Teachable Machine: The system will generate standardized claim Summary card and use Google Teachable Machine to independently classify claims and provide confidence scores.

Model Comparison & Consistency: System will compare Python and Teachable Machine predictions, calculate confidence differences and determine the level of agreement between both models.

Warranty Rule Validation: It verifies warranty conditions such as expiry, fault coverage and required documents.

Alerts & Monitoring: System will provide warranty-expiry notifications and monitor unusual activities, low confidence predictions and model disagreements.

Reporting & Data Export: System will generate downloadable claim reports and allow authorized administrators to export claim and analytical data in CSV file.

Database & Audit Trail: It will securely store users, products, warranties. Claims, documents, predictions and audit records while maintaining model versions and important users/ system activities.

Error Handling & User Interface: The system will provide understandable and messages, user-friendly web interface accessible through desktop, tablet and mobile devices.

NON-FUNCTIONAL REQUIREMENTS:

Accuracy: System will accurately analyze claim data and provide reliable claim decisions.

Security: System should protect customer and claim information.

Reliability: System should operate consistently and minimize system failures.

Maintainability: System should be easy to update, debug and maintain when new features are introduced.

Compatibility: The system should work properly across commonly used browsers and devices.

APPLICATION ARCHITECTURE:

Presentation Layer: Provides the user interface for claim submission, document upload, claim tracking and viewing claim decisions.

Application Layer: Handles claim validation and communication between different system components.

AI/ML Layer: Uses OCR, NLP, ML and decision tree models to extract information.

Data Layer: Stores customer details and claim decisions in a secure database.

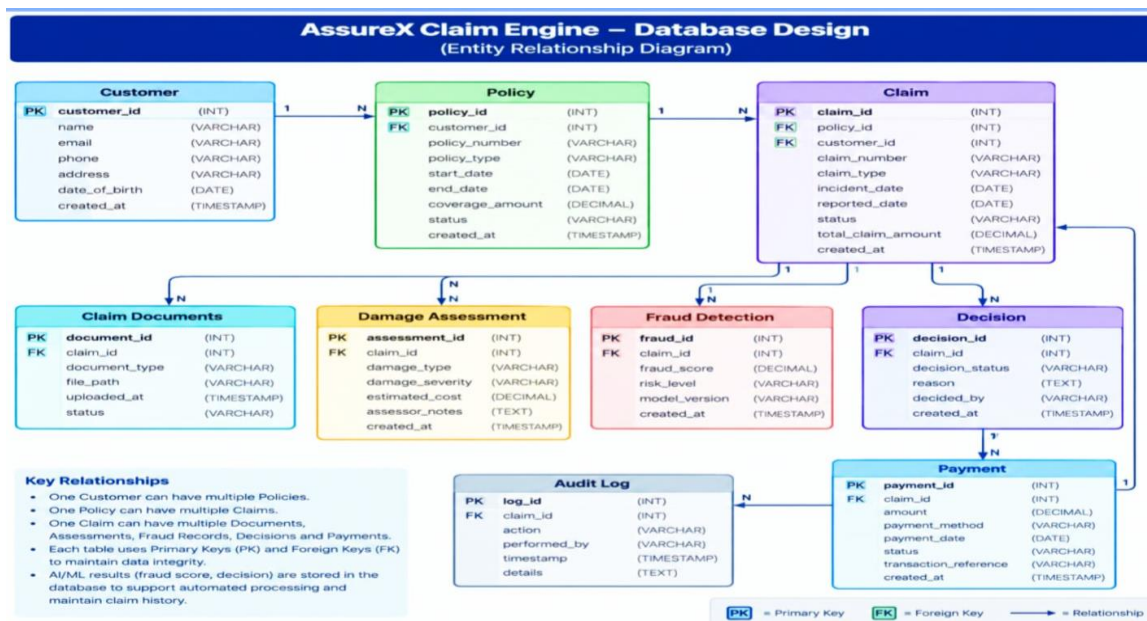
Integration Layer: Allows the claim engine to communicate with external systems.

MODULE DESCRIPTIONS:

AssureX Claim engine consists of different modules:

Management Module, Submission Module, Document Processing Module, The Claim Validation Module, AI/ML Module, Database Module, Notification Module and Reporting Module.

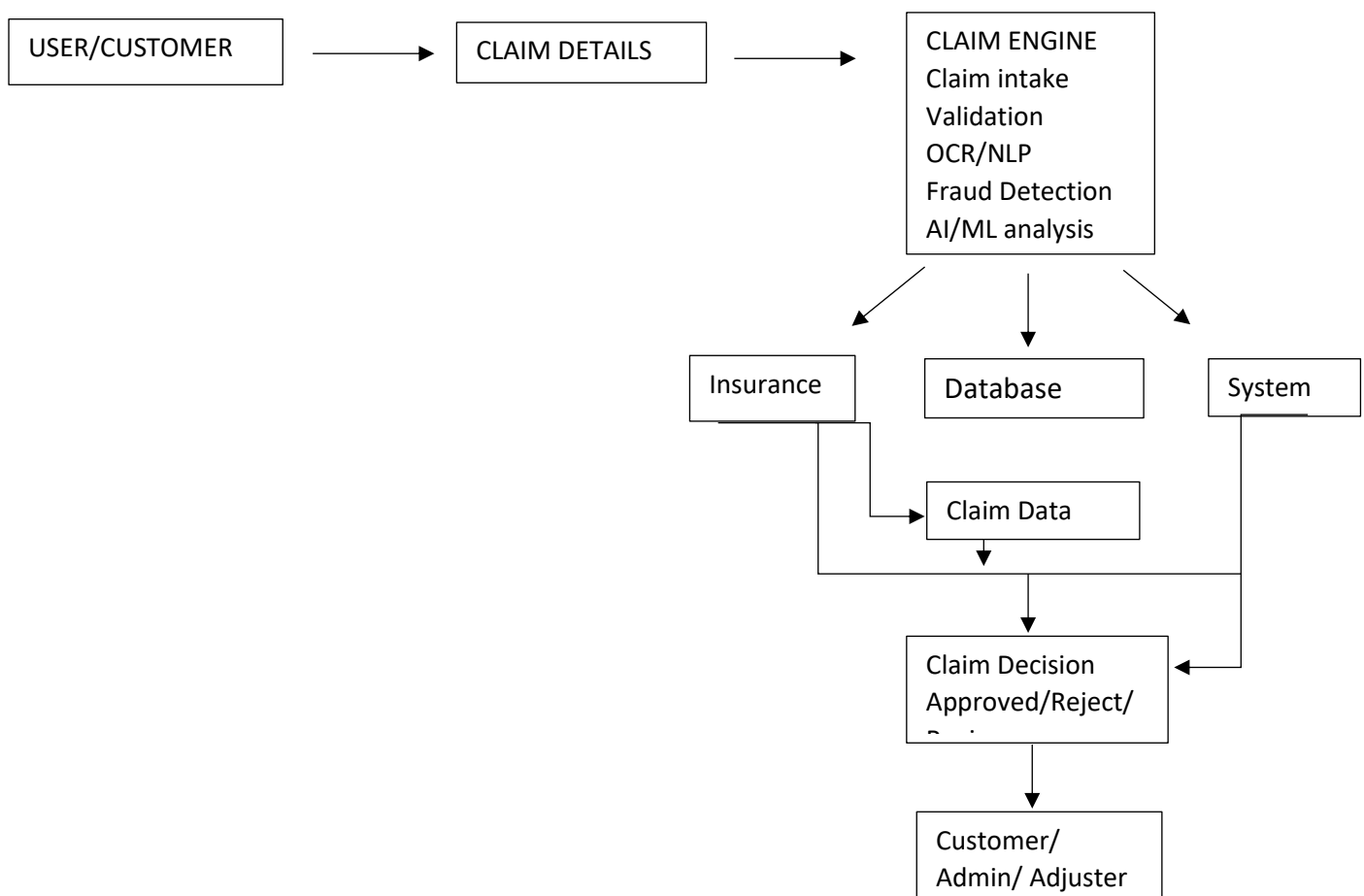
DATABASE DESIGN:



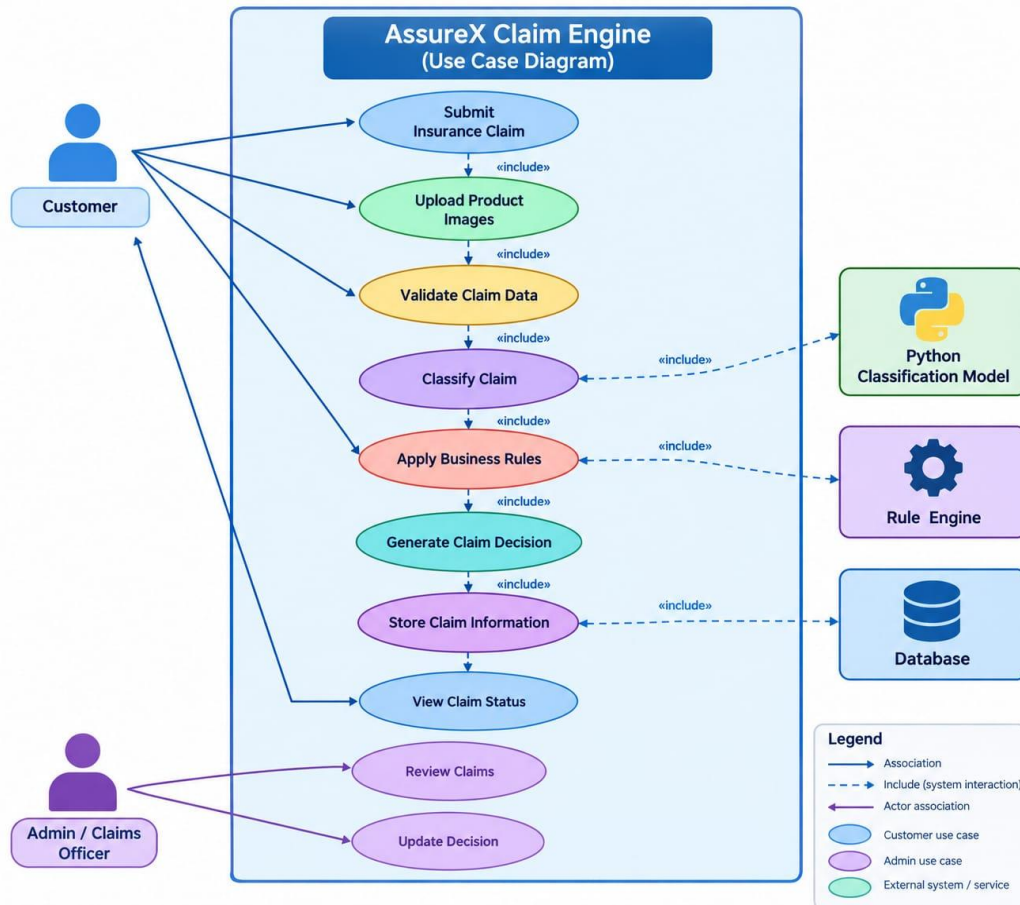
DATA DICTIONARY:

A data dictionary is a structured description of all important data fields used in AssureX claim Engine. Field names are product ID, Purchase Date, Claim date.

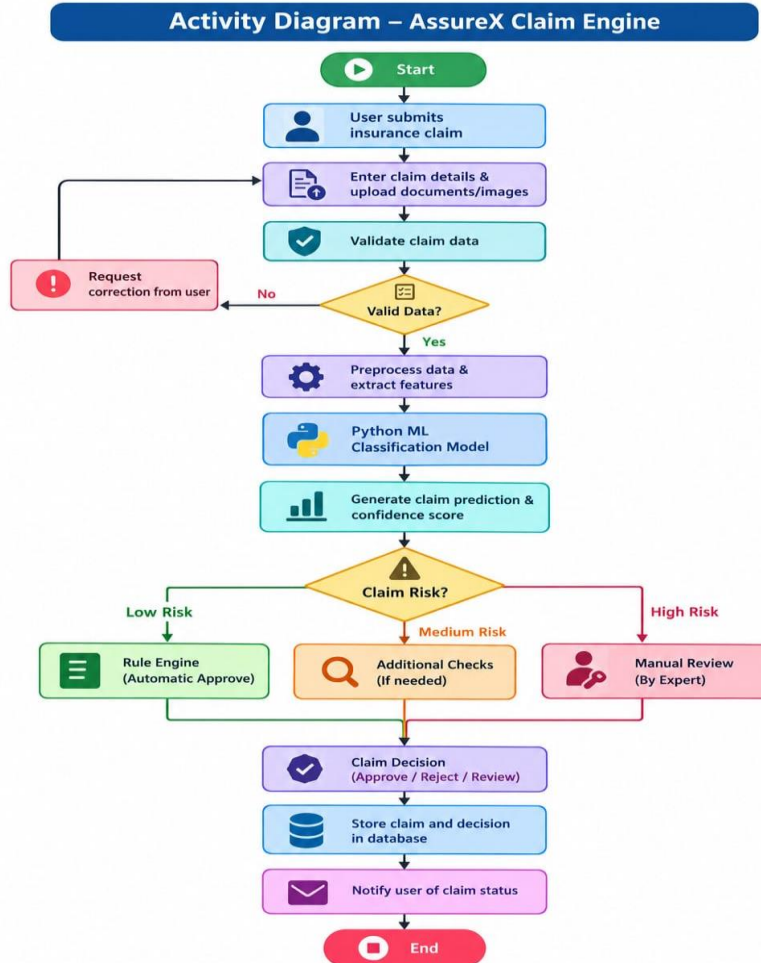
DATA FLOW DIAGRAM:



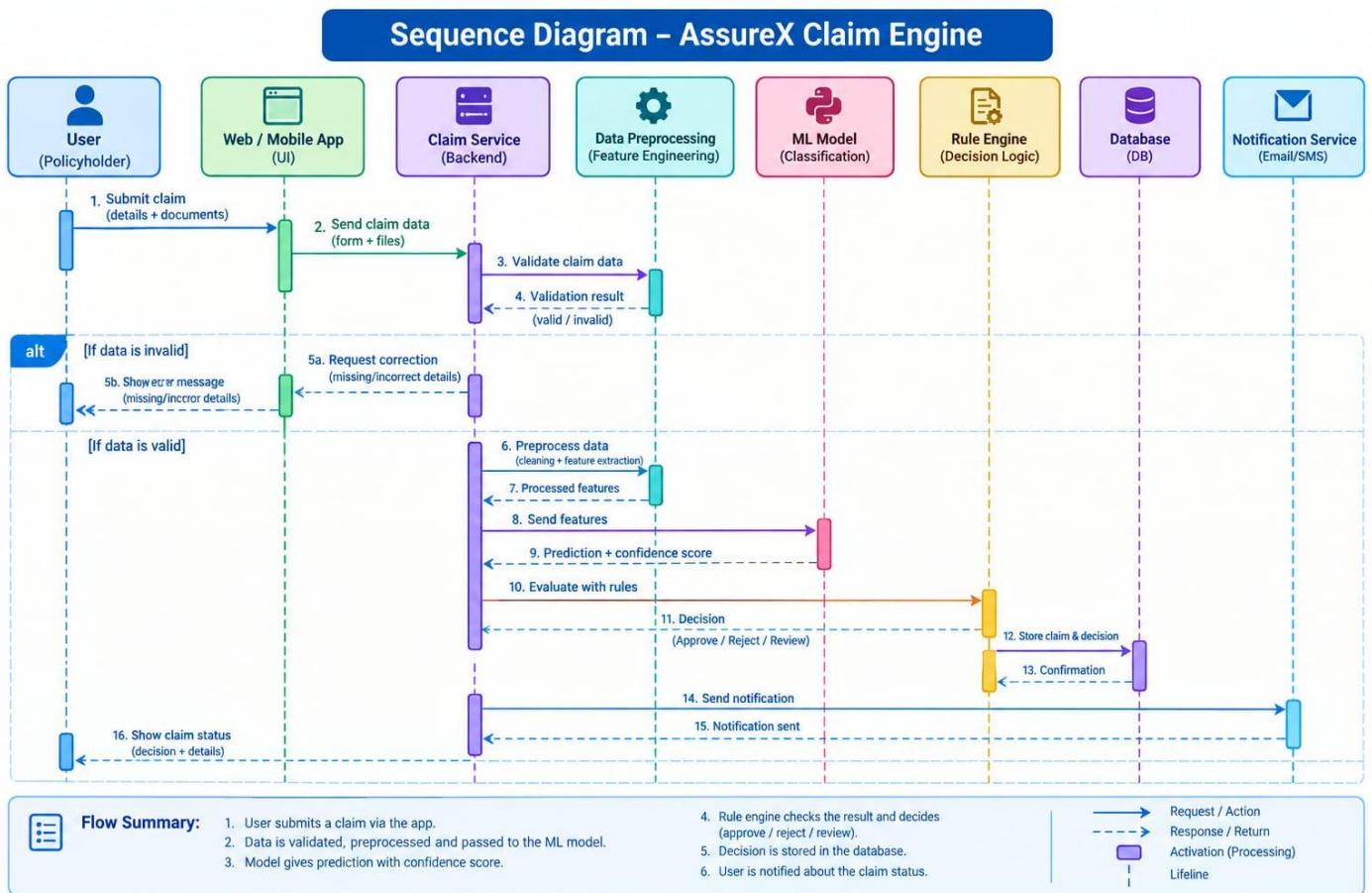
USE CASE DIAGRAM:



ACTIVITY DIAGRAM:

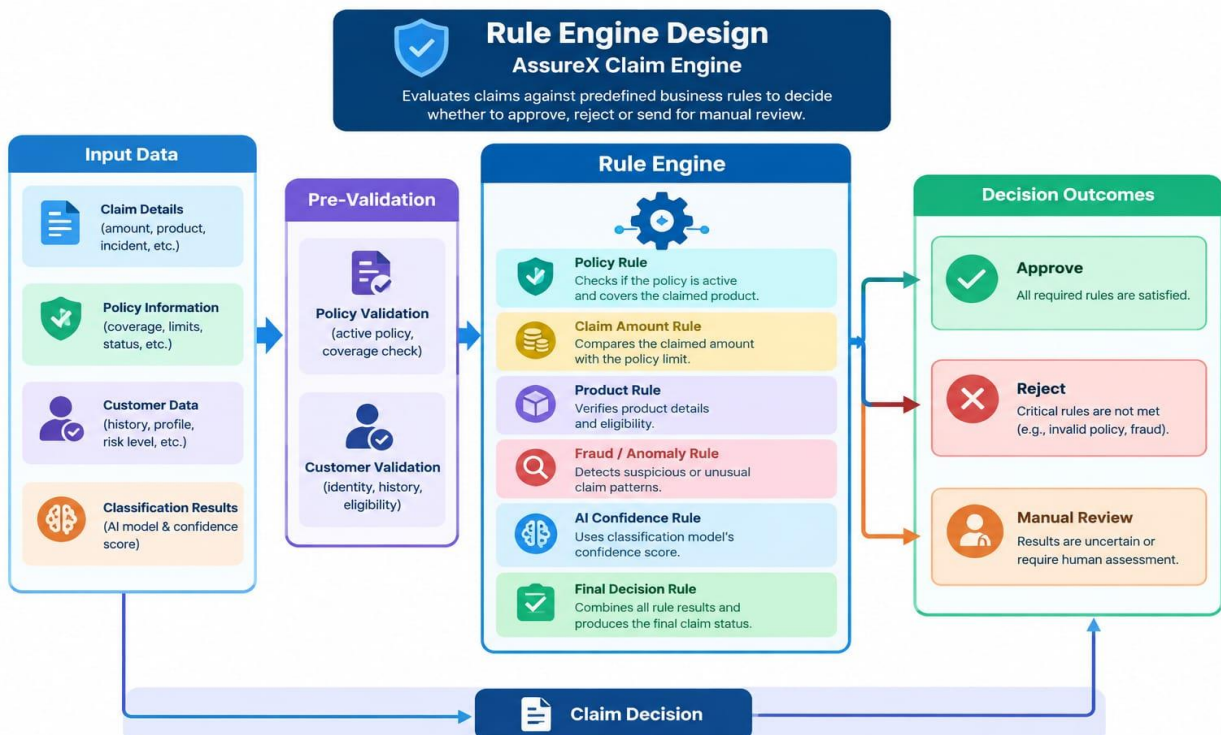


SEQUENCE DIAGRAM:



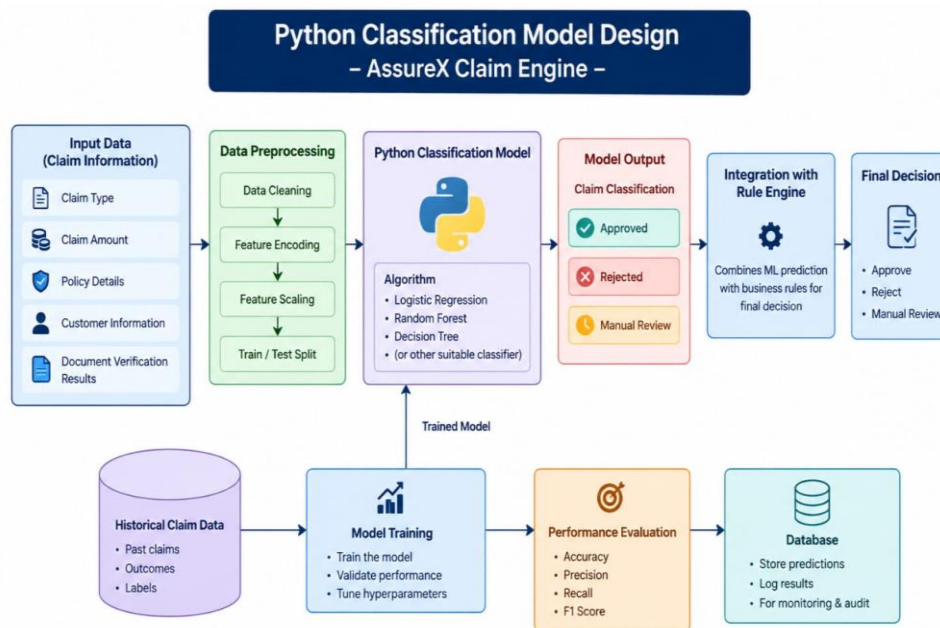
RULE ENGINE DESIGN:

The Rule Engine is designed as a decision-making component between claim validation and claim processing. It generates a decision such as Approved, Rejected or Manual Review. All rule results and decisions are stored in the database for tracking and auditing.



PYTHON CLASSIFICATION MODEL DESIGN:

The python classification model in the AssureX Claim Engine is designed to classify insurance claims based on their provided data.



GOOGLE TEACHABLE MACHINE MODEL DESIGN:

In the AssureX claim Engine, it can be used to classify claim-related images or documents into predefined categories. The model produces a classification result with a probability score, which can be used by the claim engine for further processing.

Google Teachable Machine – Model Design (For Electronic Product Classification)

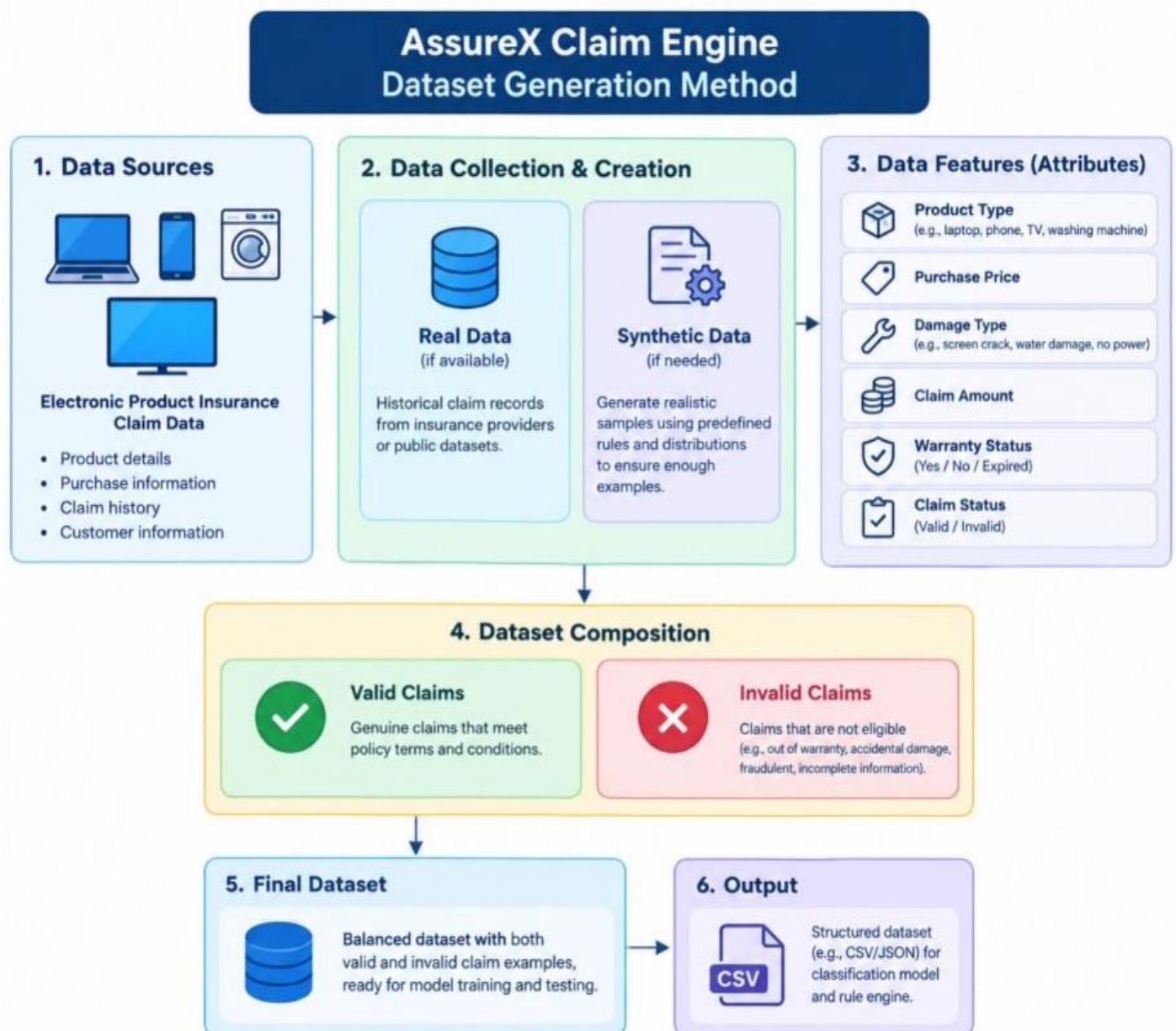


DATASET DESCRIPTION:

It contains electronic product insurance claim information used to train and test the claim classification model. It may include details such as product ID, Damage type, claim amount, warranty status and claim history. The data is based on predefined rules and patterns this data is used by the classification model.

DATASET GENERATION METHOD:

The dataset is generated by collecting and creating sample electronic product insurance claim data. Both valid and invalid claim examples are included to train and test the classification model and rule engine.



DATA PROCESSING AND FEATURE-ENGINEERING STEPS:

In this step data is cleaned, organized, transformed and important features are extracted so that the machine learning model can accurately classify insurance claims.

- Data Collection
- Data cleaning
- Data transforming
- Feature Extraction
- Feature Selection
- Data splitting

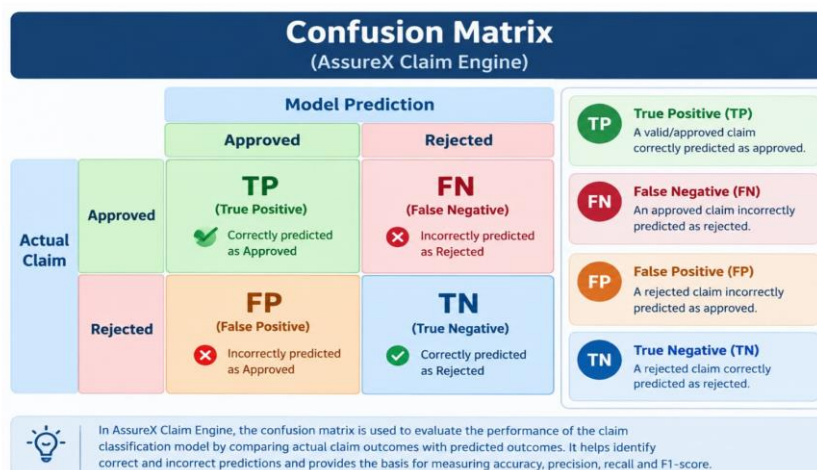
PYTHON MODEL- TRAINING PROCEDURES:

Python is used to train the classification model on the processed insurance claim dataset. The procedure starts by loading the dataset, preprocessing the data, selecting important features and splitting the data into training and testing sets.

- Load dataset
- Preprocess data
- Feature selection
- Split Dataset
- Train Model
- Evaluate Model
- Tune Model
- Save Model

CONFUSION MATRIX:

It is an evaluation method used to check how accurately the AssureX Claim Engine classification model predicts insurance claims. It helps identify correct and incorrect predictions and provides the basis for measuring accuracy, precision and recall.



ACCURACY, PRECISION, RECALL AND F1-SCORE:

Accuracy: It measures how many predictions made by the model are correct out of all predictions.

Formula: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$

Precision: It measures how many of the claims predicted as a particular class by the model are actually correct. It is useful for checking the reliability of positive predictions.

Formula: $\text{Precision} = TP / (TP + FP)$

Recall: It shows how well the model finds relevant cases.

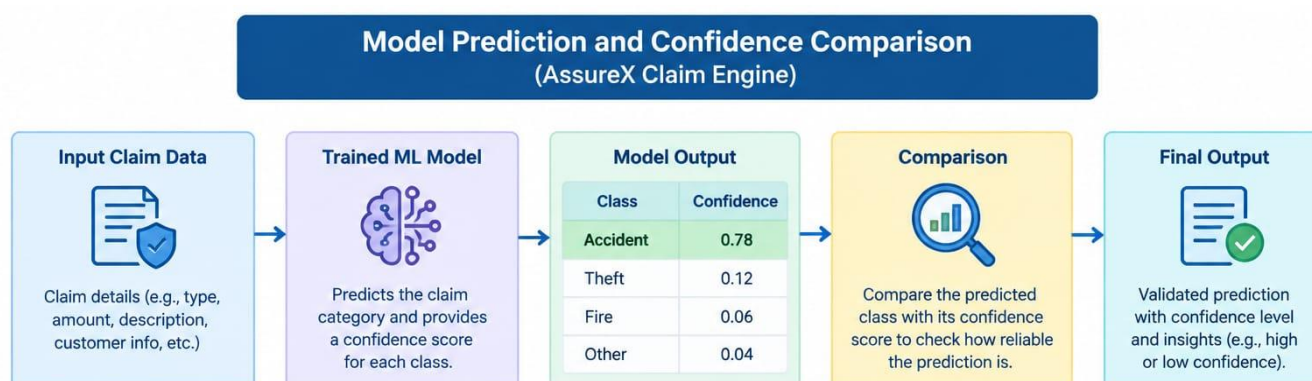
Formula: $\text{Recall} = TP / (TP + FN)$

F1-Score: It is useful when both false positives and false negatives are important in the claim classification process.

Formula: $\text{F1-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

MODEL PREDICTION AND CONFIDENCE COMPARISON:

The model predicts the category of an insurance claim, while the confidence score shows how certain the model is about that prediction. This comparison helps evaluate the reliability of the model's decisions and identify predictions where the model is less confident.



Example: Model Prediction and Confidence Comparison

Claim ID	Actual Class (For Reference)	Predicted Class	Confidence Score (of Predicted Class)	Comparison Result
C001	Accident	Accident	0.78	✓ Correct & High Confidence
C002	Theft	Theft	0.65	✓ Correct & Medium Confidence
C003	Fire	Accident	0.42	✗ Incorrect & Low Confidence
C004	Other	Other	0.91	✓ Correct & Very High Confidence

What the Comparison Tells Us

- ✓ Shows if the model's prediction is correct or incorrect.
- 📊 Helps identify predictions with high or low confidence.
- ⚠️ Highlights cases where the model is less sure (need for review or additional checks).
- ⚙️ Improves the reliability and trust in the model's decisions.



Definition

Model Prediction and Confidence Comparison is the process of comparing the class predicted by the machine learning model with its confidence score. The **model predicts** the category of an insurance claim, while the confidence score shows how certain the model is about that prediction. This comparison helps evaluate the reliability of the model's decisions and identify predictions where the model is less confident.

TESTING STRATEGY:

The system can be tested at different levels, including unit testing and system testing. Testing ensures that the claim engine gives correct classifications and handles invalid or unexpected claim data properly.

SECURITY CONSIDERATION:

Security considerations focus on protecting sensitive insurance and customer information while ensuring that the claim classification system remains reliable.

- Data Privacy
- Authentication and Authorization
- Data Encryption
- Secure Database
- Input validation
- API Security
- Model Security
- Audits logs
- Backup and recovery
- Error Handling

PRIVACY CONSIDERATION:

AssureX Claim Engine handles sensitive customer and insurance claim information, so privacy must be maintained throughout the system. The system should use authentication and role-based access control to prevent unauthorized access. Claim images and documents should also be protected from unauthorized downloading.

LIMITATIONS:

The system has some limitations. The accuracy of the classification model depends on the quality and size of the available dataset. The system may also have difficulty handling unusual or complex claim cases that are not included in the training data. The system also requires proper security and reliable data input to produce accurate and consistent results.

FUTURE ENHANCEMENT:

In the future, advanced AI and deep learning techniques can be integrated to handle complex claim cases and improve prediction confidence. The system can also be enhanced with automated fraud detection. Real-time claim processing and integrated with external insurance databases. A mobile or web-designed dashboard can further help users and insurance staff monitor claims, predictions and decision efficiently.